

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill is faintly visible in the background.

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THERAPEUTIC DRUG MONITORING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

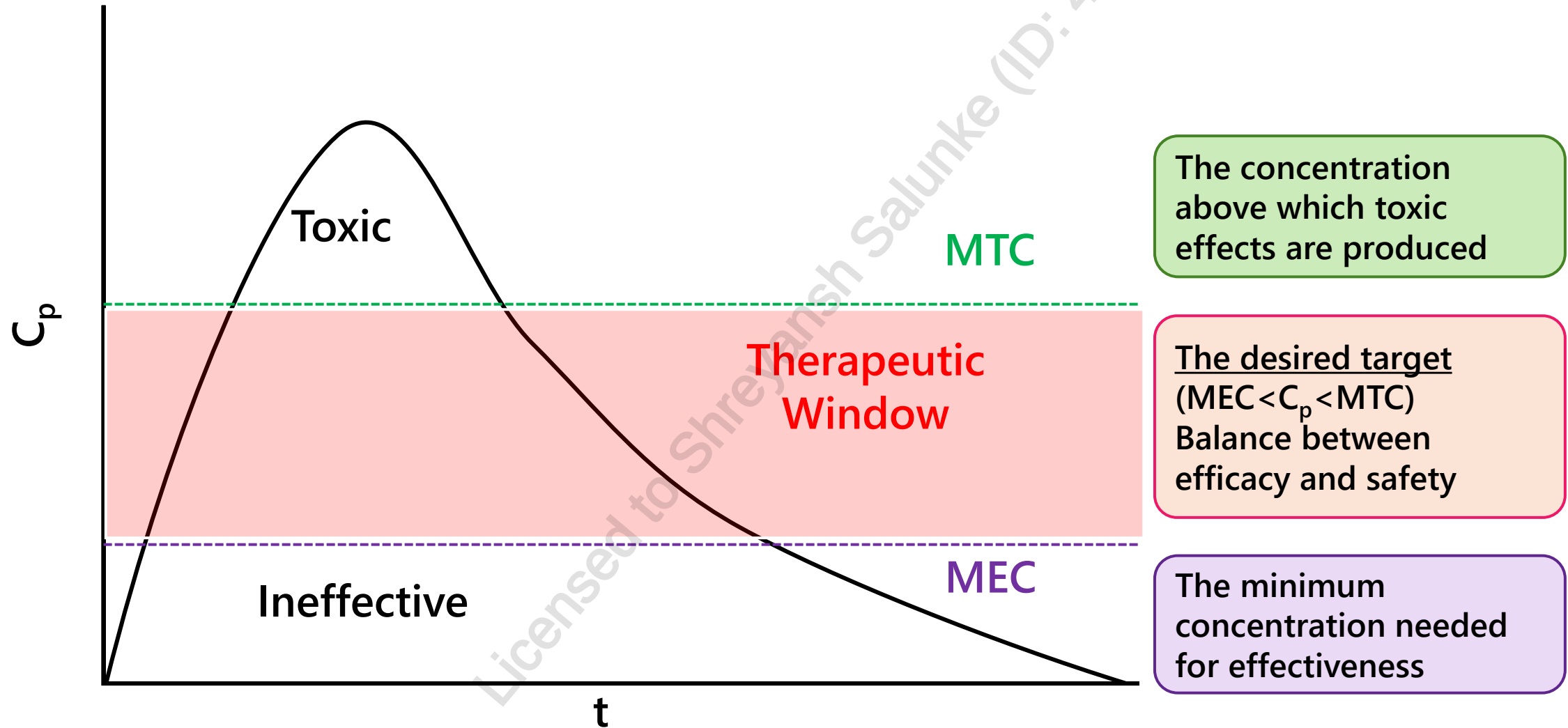
Professionalism

LEGEND

AUC	Area Under the Curve
BID	Bis In Die (Twice Daily)
BSA	Body Surface Area
CF	Cystic Fibrosis
CKD	Chronic Kidney Disease
CNS	Central Nervous System
CrCl	Creatinine Clearance
Cp	Plasma Concentration
COPD	Chronic Obstructive Pulmonary Disease
CYP	Cytochrome P450
EID	Extended-Interval Dosing
F	Bioavailability
f_u	Free Fraction
GI	Gastrointestinal
H	Hours
HFrEF	Heart Failure with Reduced Ejection Fraction
IBW	Ideal Body Weight
IM	Intramuscular
IV	Intravenous
IV	Intravenous

MEC	Minimum Effective Concentration
MIC	Minimum Inhibitory Concentration
MRSA	Methicillin-Resistant Staphylococcus Aureus
MTC	Minimum Toxic Concentration
PAE	Post-Antibiotic Effect
PK	Pharmacokinetics
Pgp	P Glycoprotein
PO	Per Os (Oral)
PRN	Pro Re Nata (As Needed)
Q	Quaque (Every)
Rxn	Reaction
SCr	Serum Creatinine
SJW	St John's Wort
TBW	Total Body Weight
TEN	Toxic Epidermal Necrolysis
TDM	Therapeutic Drug Monitoring
UTI	Urinary Tract Infection
V_d	Volume of Distribution
VIT	Vancomycin Induced Thrombocytopenia

THERAPEUTIC WINDOW



LINEAR KINETICS VS. NON-LINEAR KINETICS

	Linear	Non-Linear (Saturable)
Dose-Concentration	Concentration increases proportionally with increasing dose	Concentration increases disproportionately with increasing dose
Clearance	Constant	Decreases with increasing concentration
Half-life	Constant	Depends on concentration
Examples	Most drugs	E.g. phenytoin, ethanol
Therapeutic drug monitoring	Often not required	Required



In non-linear kinetics, small dose changes can result in large and unpredictable changes in drug plasma concentrations – hence the need for TDM!

INDICATIONS FOR TDM

- Non-linear kinetics
- Narrow therapeutic index/window
- Patient variability (e.g. special populations)
- Unclear clinical response
- Toxicity risk monitoring
- Drug interactions
- Compliance monitoring

PHENYTOIN

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OVERVIEW OF PHENYTOIN

Characteristics	
Mechanism of Action	Anticonvulsant; increases efflux and decreases influx of Na ⁺ in cell membranes of the motor cortex during generation of nerve impulses
Bioavailability	~90% (oral); varies among different dosage forms
Volume of Distribution	0.6-0.7 L/kg (adults/children)
Protein binding	92% bound to plasma albumin
Metabolism	Hepatic (CYP2C9/2C19); Michaelis-Menten kinetics
Half-life	Highly variable (i.e., 10-95 hours)

SWITCHING BETWEEN PHENYTOIN FORMULATIONS: WHAT YOU NEED TO KNOW

Phenytoin formulations – not all equal!

Available formulations:

Oral tablets, suspensions, and extended-release capsules

Intravenous (phenytoin sodium)

Two chemical forms: phenytoin acid and phenytoin sodium

Formulation	Drug Form	Dose Strength (approximate)
Oral tablets (chewable), suspension	Phenytoin acid	92 mg (equivalent to 100 mg sodium)
Oral capsules, IV	Phenytoin sodium	100 mg

Key Conversion Point!

- Phenytoin sodium 100 mg = phenytoin acid 92 mg
- Free acid form contains 8% more active drug



Clinical Pearl: always monitor levels and dose adjust when switching formulations – differences can lead to toxicity or loss of seizure control

SWITCHING BETWEEN PHENYTOIN FORMULATIONS: PRACTICE PROBLEM

A patient with epilepsy has been taking phenytoin extended-release capsules 300mg PO once daily. They have requested to switch to phenytoin suspension due to swallowing difficulties. Which of the following would be the equivalent daily dose of phenytoin suspension?

- A) 300 mg
- B) 276 mg
- C) 325 mg
- D) 250 mg

Licensed to Shreyansh Salunkhe (D-4477)

HOW TO INTERPRET PHENYTOIN LEVELS: TOTAL VS. FREE

- Phenytoin is ~90% bound to albumin
 - Remember! Only unbound or free drug reflects pharmacologically active drug
- In healthy adults, free fraction is approximately 10% ($f_u = 0.1$)
- Total phenytoin (or phenytoin_{observed}) = unbound + bound

Sheiner-Tozer Equation

(for corrected phenytoin levels at 37°C)

$$C_{corrected} = \frac{C_{measured}}{(0.2 \times Albumin) + 0.1}$$

Note: use corrected phenytoin level when patient has hypoalbuminemia or renal dysfunction

Measure type	Normal range
Total phenytoin	10-20 mg/L
Free phenytoin	1-2 mg/L

WHAT CAN AFFECT PROTEIN BINDING OF PHENYTOIN?

Reduced albumin (i.e. decreased binding capacity)

- Hypoalbuminemia due to burns, hepatic cirrhosis, nephrotic syndrome, pregnancy, cystic fibrosis

Competitive Displacement

- Uremia (renal failure) – uremic toxins can displace phenytoin
- Drug-drug interactions
 - Valproic acid
 - Salicylates
 - Warfarin

Elevated Bilirubin

- Hyperbilirubinemia – can compete for albumin binding sites



Clinical Pearl: Be vigilant for these factors as they can increase free phenytoin – monitor levels or calculate the corrected concentrations

PRACTICE PROBLEM 1

A 68-year-old male (weight: 70 kg) with a history of alcohol-associated cirrhosis and hypoalbuminemia (albumin 25 g/L) presents with status epilepticus. He is phenytoin-naïve. You are asked to calculate an appropriate loading dose to attain a target phenytoin level of 15 mg/L. His serum creatinine is 100 $\mu\text{mol/L}$. Available formulations: phenytoin sodium 50 mg/mL IV solution and phenytoin acid 100 mg PO capsules. Assume V_d of 0.65 L/kg.

1. What is the most appropriate IV loading dose of phenytoin sodium?
 - a) 683 mg
 - b) 700 mg
 - c) 742 mg
 - d) 800 mg
2. Which infusion schedule is the most appropriate?
 - a) 742 mg over 10 minutes
 - b) 742 mg over 15 minutes
 - c) 742 mg over 30 minutes
 - d) 742 mg over 5 mins



PRACTICE PROBLEM 1

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3. After the initial IV loading dose, a phenytoin level was drawn. The measured phenytoin level is 9 mg/L. What is the corrected phenytoin level and is an additional loading dose required to reach the target of 15 mg/L?
- a) 12 mg/L, yes
 - b) 15 mg/L, no
 - c) 16.7 mg/L, no
 - d) 18 mg/L, no



ANTIBIOTICS

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OVERVIEW OF VANCOMYCIN

Mechanism of action	Inhibits bacterial cell wall synthesis via disruption of transglycosylation
Bacterial activity	Gram-positive organisms (e.g. MRSA)
Pharmacokinetics	• 2-compartment model • Poor oral bioavailability (100% IV bioavailability) • Volume of distribution: ~0.7 L/kg • Mostly renally cleared
Dosing	Oral: 500-2000 mg/day in 3 or 4 divided doses IV: 15-20 mg/kg/dose Q8-12H
Adverse effects	• Nephrotoxicity • Ototoxicity (rare) • Vancomycin-induced thrombocytopenia (VIT) • Erythematous rash on face/upper body (Vancomycin Infusion Rxn) • Injection site reactions (e.g., thrombophlebitis)
Therapeutic monitoring	Most accurate and optimal → AUC-guided monitoring Most practical → trough serum concentrations

DOSING – DETERMINE WEIGHT PARAMETERS

- Determine the patient's weight parameters
 - Determine ideal body weight (IBW), recall:
 - Males: $2.3 \times (\text{\# of inches above 5 ft in height}) + 50$
 - Females: $2.3 \times (\text{\# of inches above 5 ft in height}) + 45.5$

If total or actual body weight is...	Use
>120%* above IBW	Adjusted body weight ($0.4 \times (\text{TBW} - \text{IBW}) + \text{IBW}$)
Less than IBW	Total body weight
Otherwise...	Ideal body weight

* Some resources use a 130% threshold

DOSING ADJUSTMENTS IN RENAL IMPAIRMENT

- Determine CrCl using the appropriate weight parameter using Cockcroft-Gault equation
 - Note: patients may have a SCr that is falsely low
 - Malnourished
 - Bedbound
 - Elderly (≥ 65 years)
 - Taking chronic steroids
 - Chronic diseases (e.g., muscle wasting diseases)
 - Monitor renal function closely

Recommended interval adjustment for IV vancomycin in renal impairment

Creatinine Clearance (CrCl)	Dosage Interval
≥ 50 mL/min	Q12-24H
10–49 mL/min	Q24-96H
<10 mL/min	Q4-7 days

MONITORING: TROUGH VS. AUC/MIC

- Traditionally relied on trough levels (10–20 mg/L), but AUC-guided dosing is now preferred due to stronger correlation with efficacy and reduced nephrotoxicity

	Trough	AUC/MIC
When to use?	Acceptable to use when AUC-guided dosing is not feasible	All patients (including obese, critically-ill patients, pediatrics, and patients with kidney dysfunction)
Target	10-20 mg/L (for complicated, severe, or deep-seated infections: 15-20 mg/L)	400-600 mg*h/L
Advantages	- Simple, widely used - No software or special training required	- Optimal efficacy for treating <i>S. aureus</i> infections - Lower nephrotoxicity risk - Obtained 1-2 hours post-infusion using 2 levels (trough and peak)
Disadvantages	- Must wait until 4th dose to measure trough - Higher risk of nephrotoxicity with higher troughs	More resource-intensive (e.g. training, software, etc.)

PRACTICE PROBLEM 2

A 72-year-old male (80 kg, 175 cm) with a history of hypertension, type 2 diabetes, and stage 3 chronic kidney disease (CKD) is admitted to hospital with methicillin-resistant *Staphylococcus aureus* (MRSA) bacteremia. He is empirically started on vancomycin 1 g IV every 12 hours. His baseline serum creatinine is 120 $\mu\text{mol/L}$ and his renal function has been stable throughout his admission. On day 3 of therapy, a vancomycin trough level is drawn immediately before the 4th dose and reported as 22 mg/L. The institution still uses trough-based monitoring. What would be the most appropriate next step?

- a) Continue current dose; level within target range
- b) Decrease dose or extend interval due to risk of nephrotoxicity
- c) Increase dose; trough level below target range
- d) Switch to oral vancomycin for systemic infection



PRACTICE PROBLEM 3

A 64-year-old female (weight: 60 kg, height: 162 cm) who was initially admitted to the emergency department for a chronic obstructive pulmonary disease (COPD) exacerbation, now presents with signs and symptoms of hospital-acquired pneumonia. She has a past medical history significant for COPD, rheumatoid arthritis, and stage 2 chronic kidney disease (CKD) (SCr: 85 $\mu\text{mol/L}$). Her current medications include methotrexate 15 mg PO weekly, folic acid 1 mg PO daily (except on the day methotrexate is taken), salbutamol 200 mcg 1 inhalation Q4-6H PRN, Duaklir Genuair[®] (aclidinium 400 mcg/formoterol 12 mcg) 1 inhalation BID, lisinopril 5 mg PO daily, and Synjardy[®] (empagliflozin 5 mg/metformin) 500 mg PO BID. She is started on vancomycin 1.25 g IV every 12 hours as part of empiric coverage. On day 4 of therapy, a vancomycin peak level (1 hour after a 2-hour infusion) is recorded at 42 mg/L and a trough level (30 minutes before the next dose) is recorded as 19 mg/L. The hospital uses AUC-based monitoring with a target AUC_{24} of 400–600 mg·h/L.

1. What is the patient's current AUC_{24} estimated to be? (Assume first-order kinetics)
 - a) 421 mg·h/L
 - b) 555 mg·h/L
 - c) 706 mg·h/L
 - d) 842 mg·h/L

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2. As the pharmacist, what dosing adjustment would you recommend to bring the patient's vancomycin AUC_{24} into target range?
- a) None, continue with current regimen (1.25 g IV Q12H)
 - b) Change to 1 g IV Q12H
 - c) Change to 0.75 g IV Q12H
 - d) Change to 1 g IV Q24H

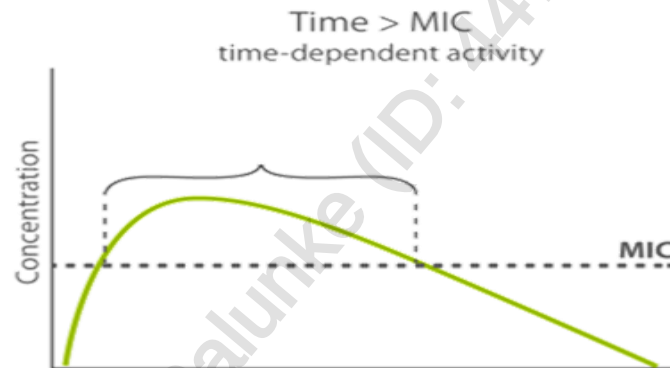
OVERVIEW OF GENTAMICIN

Mechanism of action	Inhibit bacterial protein synthesis through irreversible binding to the 30S bacterial ribosomal subunit
Bacterial activity	Gram-negative bacteria, used in gram-positive bacterial infections for its synergy with cell-wall active agents (e.g. penicillin, vancomycin)
Pharmacokinetics	• Concentration dependent killing, post-antibiotic effect • Poor oral bioavailability (IV/IM only) • Volume of distribution: 0.2-0.3 L/kg • Almost exclusively renally cleared (~99%)
Dosing	Extended-interval dosing: 4-5 mg/kg once daily Conventional dosing: 2 mg/kg x 1 dose then 1.7 mg/kg Q8H
Adverse effects	• CNS: tinnitus, headache, dizziness, ototoxicity • Renal: nephrotoxicity
Therapeutic monitoring	Extended-interval: use of dosing nomogram, troughs may be required Conventional: peak and trough recommended

ANTIBACTERIALS – PK/PD SUMMARY

Time-dependent killing

- Bacterial killing (rate and extent) determined by **duration of time** that a drug's concentration at site of infection remains **above MIC**

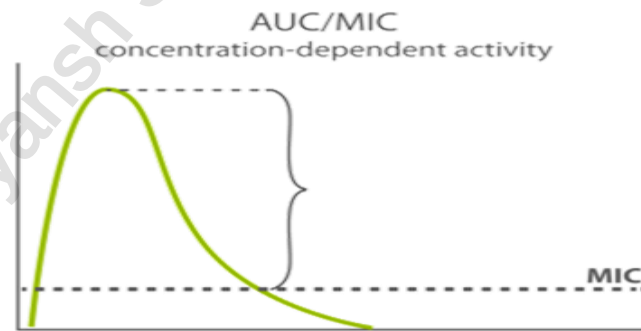


Common Examples:

- Beta Lactams
- Macrolides
- Vancomycin

Concentration-dependent killing

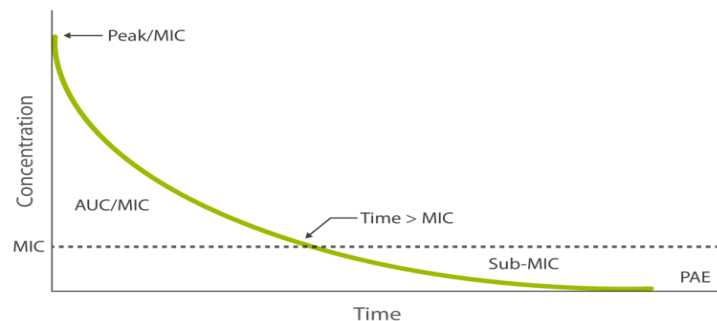
- Bacterial killing (rate and extent) determined by **peak drug levels** or AUC/MIC ratio (i.e. exposure)



- Quinolones
- Aminoglycosides

Post-antibiotic effect (PAE)

- Continued suppression of normal growth with antimicrobial levels **below the MIC**



- Aminoglycosides

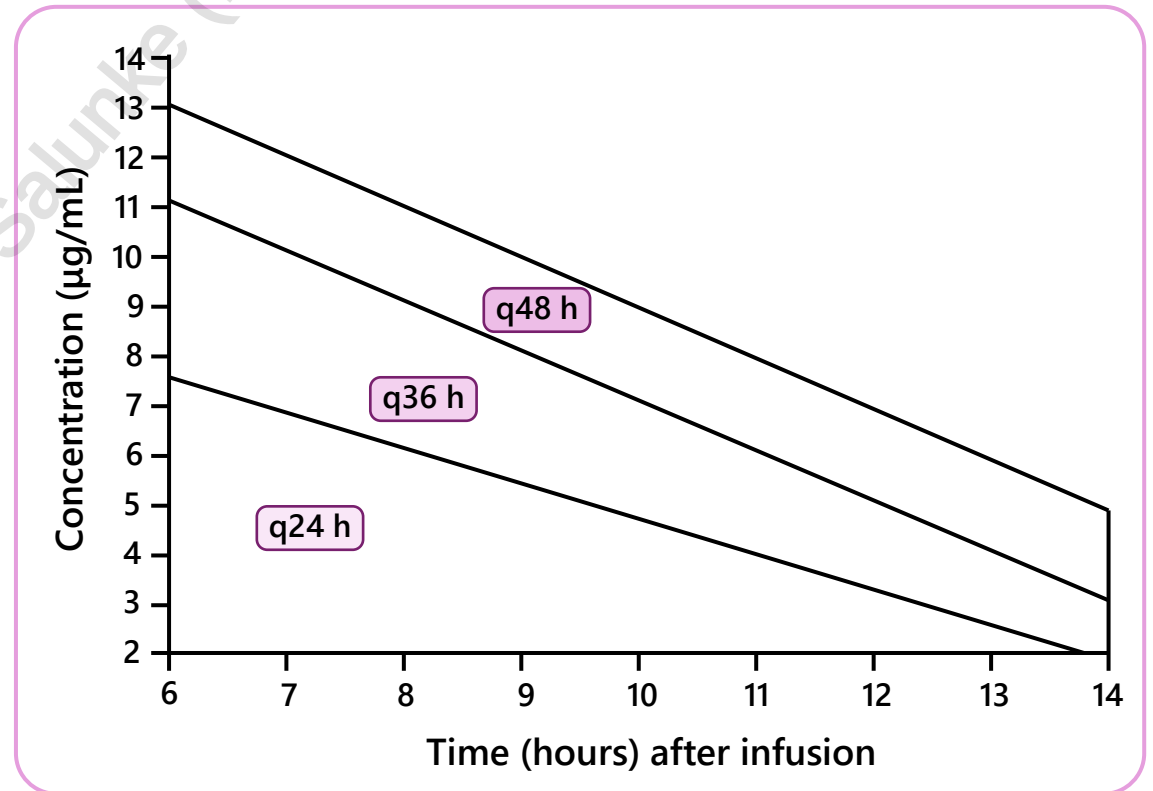
EXTENDED VS. CONVENTIONAL DOSING

	Extended-Interval Dosing (EID)	Conventional / Multiple
Definition	Once-daily, high-dose, or non-traditional dosing	Total daily dose divided throughout the day (e.g. Q8-12H)
Rationale	• Maximizes concentration-dependent killing • Drug-free interval → PAE • Adaptive resistance	• For patients with altered PK; allows for closer control • When EID is not appropriate (e.g., burns >20% of patient's BSA, CF, pregnancy, hemodialysis)
Dosing	4-5 mg/kg (7mg/kg if critically ill) Q24H	Loading: 2 mg/kg (IBW) Maintenance: 1.7 mg/kg (IBW) Q8H
TDM	• Hartford nomogram and single serum concentration	• Obtain after 3-5 half lives, or at steady-state • Peak: 4-10 mcg/mL • Trough: <1-2 mcg/mL

PRACTICE PROBLEM 4

A 58-year-old male (weight: 85 kg, height: 178 cm) with a history of type 2 diabetes mellitus and diabetic foot ulcers is admitted with complicated urinary tract infection (UTI) with bacteremia caused by *P. aeruginosa*. Gentamicin is started for Gram-negative coverage using extended-interval dosing. On day 1, he receives 7 mg/kg IV over 30 minutes. A random gentamicin level, drawn 8 hours after the start of infusion, is 4.2 mg/L. Serum creatinine = 82 $\mu\text{mol/L}$.

- Using the Hartford nomogram on the right for gentamicin at a dose of 7 mg/kg, what dosing interval is recommended for this patient based on the 8-hour post-dose level?
 - Q24H
 - Q36H
 - Q48H
 - Q60H



DIGOXIN

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OVERVIEW OF DIGOXIN

Mechanism of action	• Inhibits the $\text{Na}^+\text{-K}^+$ exchange in the $\text{Na}^+\text{-K}^+\text{-ATPase}$ pump leading to an increase in sodium → increases contractility (+inotropy) • Slows conduction through AV node –chronotropy)
Indications	• Congestive heart failure • Atrial fibrillation
Pharmacokinetics	• 2-compartment model • F: 0.7 to ~1 • Protein binding: 30% • Volume of distribution: 7 L/kg • 15-20% metabolized, ~80% renally eliminated • Half-life: can be as long as 48H
Dosing	0.125-0.25 mg daily
Adverse effects	• GI: nausea, vomiting • Neuro: weakness, confusion, lethargy • Chronic manifestations: malaise, weakness, visual disturbances (blurred vision) • Cardiotoxicity (acute and chronic)

THERAPEUTIC MONITORING OF DIGOXIN

Key Factors

- Consider when dosing: ideal body weight, CrCl, age, comorbidities, drug interactions
- Monitor renal function closely (~80% renal elimination)

Sampling Recommendations

- Draw levels ≥ 6 H (ideally 12 H) post-dose to allow distribution from blood to tissues
- In renal failure / hemodialysis: check 12-24 H post-dose
- Due to long half-life (~48 H), wait 7–14 days after initiation or dose change to check steady state level

Therapeutic Range

- 0.6-1.2 nmol/L (recommended by Canadian Cardiovascular Society)
- Levels above 2.6 nmol/L → increased risk of digoxin toxicity

FACTORS THAT CAN AFFECT DIGOXIN LEVELS

Factor	Cause	Effect
Electrolyte imbalances	Hypokalemia ($\downarrow K^+$) Hypomagnesemia ($\downarrow Mg^{2+}$)	$\uparrow$ digoxin toxicity
	Hypercalcemia ($\uparrow Ca^{2+}$)	$\uparrow$ risk of serious arrhythmias
Thyroid function	Hypothyroidism Hyperthyroidism	$\uparrow$ digoxin levels $\downarrow$ digoxin levels
Renal function	Decreased or elderly	$\uparrow$ digoxin toxicity
Concomitant medications	Pgp inhibitors: amiodarone, verapamil, diltiazem, quinidine, macrolides, azoles	$\uparrow$ digoxin levels/toxicity
	Pgp inducers: rifampin, SJW, phenytoin, ritonavir	$\downarrow$ digoxin levels

PRACTICE PROBLEM 5

A 75-year-old female (60 kg, 160 cm) with heart failure with reduced ejection fraction (HFrEF) has been receiving digoxin 0.125 mg PO once daily for the past 3 months. She was recently started on amiodarone for newly diagnosed atrial fibrillation. She now presents to clinic with complaints of nausea and blurred vision. Her laboratory results show serum creatinine 100 $\mu\text{mol/L}$, and other bloodwork parameters are within normal limits. A digoxin level drawn 2 hours after her morning dose is reported as 2.3 nmol/L.

1. What is the most appropriate interpretation and next step?
 - a) The digoxin level is above the recommended range; hold digoxin immediately due to toxicity risk
 - b) The digoxin level is likely falsely elevated due to early sampling; repeat the level ≥ 6 –12 hours post-dose
 - c) Nausea and blurred vision are transient adverse effects; no changes required
 - d) Adverse effects are due to amiodarone; discontinue amiodarone



PRACTICE PROBLEM 5

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2. A repeat digoxin level drawn 12 hours post-dose now reports a level of 1.9 nmol/L. What is the most appropriate next step?
- a) Continue therapy
 - b) Reduce digoxin dose by 50% (e.g., to 0.0625 mg daily) and monitor closely
 - c) Hold digoxin until level <1.2 nmol/L
 - d) Stop amiodarone immediately



REFERENCES

1. Yartsev A. First order, zero order and non-linear elimination kinetics. *Deranged Physiology*. 2023. <https://derangedphysiology.com/main/cicm-primary-exam/pharmacokinetics/Chapter-337/first-order-zero-order-and-non-linear-elimination-kinetics>
2. Kang JS, Lee MH. Overview of Therapeutic Drug Monitoring. *Korean J Intern Med*. 2009;24(1):1. doi: 10.3904/kjim.2009.24.1.1.
3. MILLARES-SIPIN CA, ALAFRIS A, COHEN H. Phenytoin and Fosphenytoin. In: Cohen H. eds. *Casebook in Clinical Pharmacokinetics and Drug Dosing*. McGraw-Hill Education; 2015. <https://accesspharmacy.mhmedical.com/content.aspx?bookid=1514§ionid=88805062>.
4. CPS. Ottawa (ON): Canadian Pharmacists Association; c2016 [updated October 1, 2017]. Phenytoin [product monograph]. Available from: <http://www.e-cps.ca>.
5. RICCOBONO DE, MONTEAGUDO-CHU MO. Vancomycin. In: Cohen H. eds. *Casebook in Clinical Pharmacokinetics and Drug Dosing*. McGraw-Hill Education; 2015. <https://accesspharmacy.mhmedical.com/content.aspx?bookid=1514§ionid=88805949>.
6. NOVIASKY J, DARKO W, AKAJABOR D, MAKARON A, MILLER C, PIERCE DP, SEABURY R, WHALEN K. Digoxin. In: Cohen H. eds. *Casebook in Clinical Pharmacokinetics and Drug Dosing*. McGraw-Hill Education; 2015. <https://accesspharmacy.mhmedical.com/content.aspx?bookid=1514§ionid=88803583>.
7. SALTIEL M, TART S, KILLIUS KA, TRUESDALE TH, PERRODIN D, VERKLEEREN ND, MACEIRA E, VAUGHAN C, PLEWA AM. Aminoglycoside Pharmacokinetics. In: Cohen H. eds. *Casebook in Clinical Pharmacokinetics and Drug Dosing*. McGraw-Hill Education; 2015. <https://accesspharmacy.mhmedical.com/content.aspx?bookid=1514§ionid=88802784>.
8. McDonald M, Virani S, Chan M, Sussex B. CCS/CHFS Heart Failure Guidelines Update: Defining a New Pharmacologic Standard of Care for Heart Failure With Reduced Ejection Fraction. *Canadian Journal of Cardiology*. 2021;37(4):531-546. doi: 10.1016/j.cjca.2021.01.017.

CHANGE LOG

September 2025

- New lecture

October 2025

- Slide 20: Added note that some resources use a 130% threshold

March 2026

- Slide 2: Updated NAPRA competencies